

Hong Lang

Postdoctoral Research Associate · Civil & Environmental Engineering, University of Illinois Urbana-Champaign
Research Associate Professor · Transportation Engineering, Tongji University
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Research Summary

Physical AI for Roadway Infrastructure — from the instrument to the agency decision.

I own the full measurement stack for the road itself: I design and calibrate the 3-D imaging and LiDAR-INS systems that acquire the data, develop physics-informed learning models that convert it into the variables transportation engineers design and program with — roadway alignment and geometry, surface condition and friction, tire-pavement contact stress — and deliver those models as tools that state DOTs run in production. My systems have been deployed on **50,000+ km** of highway, my tools are in use at the Illinois Center for Transportation, and I lead a **\$653 K IDOT** project rebuilding the statewide Illinois Pavement Feedback System. Owning this chain end to end is what lets me move beyond describing a road's current condition toward **predicting how it degrades**, linking measured geometry and friction to crash risk and preservation investment, and — because I own the statewide data substrate — extending it with vision-language and agent-based interfaces so engineers can query and act on network-level asset data directly.

Focus: Mobile LiDAR & 3-D measurement systems · Roadway geometry and alignment extraction · Physics-informed learning for infrastructure performance · Network-scale pavement data platforms and asset decision-support tools · Foundation models for cross-regional infrastructure data

Record: \$1.02 M as sole PI (IDOT, NSFC, provincial DOTs) and \$2.85 M as Co-PI / senior personnel · 43 peer-reviewed journal articles (IEEE T-ITS/TIM, Automation in Construction, TRR, ASCE) · 17 TRB papers · 7 patents · 2 national standards · 15+ students mentored

Education

Tongji University, Shanghai, China 2016–2021
Ph.D. in Transportation Engineering (Highest Distinction)

- Dissertation: *Automatic Distress Detection and Classification Algorithms Using 3-D Pavement Images*
- Advisor: Dr. Jian John Lu (Dean of the College of Transportation)

Tianjin University of Science & Technology, Tianjin, China 2012–2016
B.Eng. in Measurement and Control Technology & Instrument (Highest Distinction)

Academic Appointments

University of Illinois Urbana-Champaign, Champaign, IL 2024–Present
Postdoctoral Research Associate, Illinois Center for Transportation (ICT)

- Advisor: **Dr. Imad L. Al-Qadi** (Grainger Distinguished Chair in Engineering)
- Secured **\$653 K IDOT grant as sole PI** (ICT-R27-288) for modernizing the Illinois Pavement Feedback System. <https://trid.trb.org/View/2677555>.
- Led development of AI-based pavement mix design and binder selection tools deployed by ICT.
- Mentored 6 UIUC Ph.D. students on GPR, pavement friction, WIM, and SMA-APT projects.
- Co-authored or contributed to 9 submitted proposals across FHWA, IDOT, WisDOT, MnDOT, and USDOT.

Tongji University, Shanghai, China 2024
Research Associate Professor, Transportation Research Institute

- Served as sole PI on projects funded by NSFC and the Shanghai and Henan DOTs.

Postdoctoral Research Fellow, College of Transportation

2021–2024

- Advisors: Dr. Jian John Lu & [Dr. Jinsong Qian](#) (Dean of Dept. of Transportation Infrastructure)
- Built an open-access dataset of 88,991 pavement images and 800 km MLS point clouds (GitHub: IP2RG).

Research Grants

Funding Summary — Total as PI: ~\$1.02 M | Total as Co-PI / Senior Personnel: ~\$2.85 M

As Principal Investigator

Modernization and Web-Based Implementation of the Illinois Pavement Feedback System

Illinois Department of Transportation (IDOT), ICT-R27-288 PI: Hong Lang \$653 K · Mar 2026–Feb 2028

High-Resolution 3D Runway Damage Detection for Airport Pavement Monitoring

Tongji University, P.2023-03 PI: Hong Lang \$250 K · Mar 2023–Jun 2024

Intelligent Monitoring and Analysis of Asphalt Pavement Service Condition

Henan Provincial DOT, P.2023-1-1 PI: Hong Lang \$60 K · Jan 2023–Mar 2025

Intelligent Pavement Distress Detection Using Bidirectional 3-D Triangulation and Multi-information Fusion

National Science Foundation of China (NSFC), P.62206201 PI: Hong Lang \$30 K · Jul 2022–Jul 2024

Pavement Distress Mechanism of Spatial Domain Generalization by Multi-Data Fusion

China Postdoctoral Science Foundation, P.2023M732644 PI: Hong Lang \$12 K · Jul 2023–Jul 2024

Development and Application of an AI-Based Pavement Inspection Data System

Shanghai Science & Technology Commission, P.23692118200 PI: Hong Lang \$7.5 K · Mar 2023–Mar 2024

Bridge Differential Settlement Detection Based on Inertial Navigation

Jiangsu Provincial Key Laboratory, P.TTS2021-03 PI: Hong Lang \$4.5 K · Mar 2021–Dec 2023

As Co-PI / Senior Personnel

Integrated Active Road Traffic Safety Prevention and Control Technology

National Key R&D Program of China, R2017YFC0803902 PI: Jian John Lu; Co-PI: Hong Lang \$660 K · 2018–2023

Key Technologies and Equipment for Vehicle-Mounted Lightweight and Rapid Inspection of Highway Operational Safety Incidents

Key R&D Project of Henan Province, P.261111242600 PI: Jinsong Qian; Co-PI: Hong Lang \$47 K · 2024–2025

Modeling and Optimization Control of Interactive Behaviors in Expressway Traffic under Mixed Traffic Conditions

NSFC, P.52472351 PI: Yajie Zou \$69 K · 2024–2025

Optimizing the Use of Local Aggregates in SMA Accelerated Pavement Testing

IDOT, R27-216 PI: Imad L. Al-Qadi \$560 K · Oct 2024–Dec 2025

Automated and Contactless Identification of Asphalt Pavement Surface Friction Based on Computer Vision Image-based 3D Reconstruction and Machine Learning Techniques

IDOT, ICT-R27-247 PI: Imad L. Al-Qadi; Co-PI: Mani Golparvar-Fard \$480 K · Jul 2024–Jun 2026

Using Advanced Binder Rheological Parameters to Predict Cracking Potential of Hot-Mix Asphalt Mixtures with Modified Binders

IDOT, ICT-R27-250 PI: Imad L. Al-Qadi \$540 K · Oct 2024–Sep 2025

Update of Traffic Factor Equations for IDOT Mechanistic-Empirical Pavement Design

IDOT, ICT-R27-277 PI: Imad L. Al-Qadi

\$489 K · Dec 2024–Present

Publications

Summary: 43 peer-reviewed journal articles (23 listed below; 6 in IEEE T-ITS/TIM and *Automation in Construction*) · 17 peer-reviewed conference papers · * corresponding author, † co-first author

Peer-Reviewed Journal Articles

- [J23] X.T.Chen; **H.Lang***; Z.Zou; Y.Chen; Y.M.Jiang. “Coarse-to-refined Road Alignment Extraction and Parameterization from MLS Point Clouds.” *Autom. Constr.*, 2026.
- [J22] A.D.Wang; **H.Lang***; J.Chen; Z.Zou; Y.J.Zou; J.S.Qian. “Automated Pavement Pothole Detection and Multi-dimensional Indicators Extraction Using Deep Learning and 3D Reconstruction.” *IEEE Trans. Intell. Transp. Syst.*, 2026.
- [J21] **H.Lang***; W.D.Villamil; I.L.Al-Qadi. “3D Tire–Pavement Contact Stresses: Physics-Informed Prediction Approach.” *Int. J. Pavement Eng.*, 27(1), 2026.
- [J20] A.O.Sulaiman*; **H.Lang**; I.L.Al-Qadi. “Hot-Mix Asphalt Cracking Potential Prediction from Binder Rheological Parameters.” *Transp. Res. Rec.*, 2026 (In Production).
- [J19] **H.Lang***; I.L.Al-Qadi; M.A.Uthman. “Designing Asphalt Concrete Mixtures Based on BMD Requirements Utilizing A Two-Step Approach.” *Transp. Res. Rec.*, 2025.
- [J18] **H.Lang**; Z.Chen*; J.S.Qian*; A.D.Wang; Y.C.Peng; H.Du. “Automated Image-level Pavement Type Recognition on Cross-regional Data Using a Multi-feature FusionNet.” *Int. J. Pavement Eng.*, 2025.
- [J17] Y.H.Si†; **H.Lang†**; J.S.Qian*; Z.Zou. “LiDAR-Driven Innovations in Pavement Distress Detection: A Review.” *ASCE J. Comput. Civ. Eng.*, 40(1): 03125003, 2025.
- [J16] A.D.Wang; **H.Lang***; Y.C.Peng; X.T.Chen; J.S.Qian. “Selecting and Optimizing Distress Feature Descriptors from Multi-source Images for Pavement Distress Classification.” *Int. J. Transp. Sci. Technol.*, 2025.
- [J15] H.Y.Ouyang; J.S.Qian*; **H.Lang**; J.K.Zhang; Y.Zhang. “Smartphone-Based Image Analysis Method for Estimating Moisture Content in Subgrade Filling.” *Int. J. Transp. Sci. Technol.*, 2025.
- [J14] H.L.Chen; C.Y.Xue; **H.Lang**; et al. “The Unification of Damage Evolution in Asphalt Mixtures under Various Strain Waveforms Using the Dissipated Energy Indicator.” *Constr. Build. Mater.*, 493, 143214, 2025.
- [J13] T.R.Zhang; **H.Lang**; Y.B.Chen*; A.Moeinaddini*; Y.Zou. “Optimization of Water-road Freight Transportation Routes for Reduced Fuel Consumption and Traffic Risk.” *IET Intell. Transp. Syst.*, 19(1), e70078, 2025.
- [J12] J.C.Jiang; **H.Lang**; Y.B.Chen*; et al. “A Dynamic Dual Optimization Method for Safe Lane Changing in Autonomous Driving.” *Transportmetrica A*, 1–33, 2025.
- [J11] Z.Zou; **H.Lang***; J.J.Lu. “Coarse-to-Refined Road Curb Segmentation From MLS Point Clouds.” *Autom. Constr.*, 166: 105586, 2024.
- [J10] A.D.Wang; **H.Lang***; Z.Chen; Y.C.Peng; J.J.Lu. “The Two-step Method of Pavement Pothole and Raveling Detection and Segmentation Based on Deep Learning.” *IEEE Trans. Intell. Transp. Syst.*, 2024.

- [J9] **H.Lang**; Y.Yuan*; J.Chen; S.Ding; J.J.Lu. “Augmented Road Crack Segmentation: Learning Complete Representation to Defend Noise in Concrete Pavements.” *IEEE Trans. Instrum. Meas.*, 2024.
- [J8] **H.Lang**; J.S.Qian; Y.Yuan*; J.Chen; Y.Y.Xing; A.D.Wang. “Automatic Pixel-Level Detection of Multiple Pavement Distresses and Surface Design Features with PDSNet II.” *ASCE J. Comput. Civ. Eng.*, 2024.
- [J7] Z.Zou; **H.Lang***; Y.X.Lou; J.J.Lu. “Plane-based Global Registration Method for Pavement 3D Reconstruction Using Hybrid Solid-state LiDAR.” *Autom. Constr.*, 152: 104907, 2023.
- [J6] T.Wen; S.Ding; **H.Lang***; J.J.Lu; Y.Yuan; Y.C.Peng; J.Chen; A.D.Wang. “Automated Pavement Distress Segmentation on Asphalt Surfaces Using a Deep Learning Network.” *Int. J. Pavement Eng.*, 24(2), 2027414, 2023.
- [J5] S.Ding; **H.Lang***; J.Chen; Y.Yuan; J.J.Lu. “Automated Crack Segmentation on 3D Asphalt Surfaces with Richer Attention and Hybrid Pyramid Structures.” *Int. J. Pavement Eng.*, 24(1), 2246097, 2023.
- [J4] T.Wen; **H.Lang***; S.Ding; J.J.Lu; Y.Y.Xing. “PCDNet: Seed Operation-Based Deep Learning Model for Pavement Crack Detection on 3D Asphalt Surface.” *ASCE J. Transp. Eng. Part B: Pavements*, 148(2): 04022023, 2022.
- [J3] S.Ding; Y.Y.Xing; **H.Lang**; T.Wen; J.J.Lu*. “Line-Structured Light Rut Detection of Asphalt Pavement with Markings Interference under Strong Light.” *ASCE J. Transp. Eng. Part B: Pavements*, 148(2), 04022007, 2022.
- [J2] Y.X.Lou; S.D.Chen; J.J.Lu; **H.Lang**. “A Portable Measurement System for Pavement Surface Macrotexture.” *Int. J. Pavement Eng.*, 23(12), 4136–4148, 2022.
- [J1] **H.Lang**; J.J.Lu; Y.X.Lou*; S.D.Chen. “Pavement Cracking Detection and Classification Based on 3D Image Using Multiscale Clustering Model.” *ASCE J. Comput. Civ. Eng.*, 34(5): 04020034, 2021.

Under Review / In Revision

- **H.Lang***; I.L.Al-Qadi. “Probabilistic Compliance Assessment for Balanced Mix Design.” *TRB 106th Annual Meeting of the Transportation Research Board, Washington, D.C., USA* (under review).
- M.Fakhreddine; **H.Lang**; M.Golparvar-Fard*; I.L.Al-Qadi. “Data-drive Network-level Pavement Friction Prediction.” *TRB 106th Annual Meeting of the Transportation Research Board, Washington, D.C., USA* (under review).
- A.Singh; **H.Lang**; I.L.Al-Qadi. “Prediction of Seasonal Axle-Load Spectra for Non-Weigh-in-Motion Interstates Using Parallel Factor Analysis.” *TRB 106th Annual Meeting of the Transportation Research Board, Washington, D.C., USA* (under review).
- Z.Chen; **H.Lang***; B.Yang; J.S.Qian. “Lightweight Joint Classification of Pavement Material and Distress Presence Using a Dual-Branch ConvNeXt.” *TRB 106th Annual Meeting of the Transportation Research Board, Washington, D.C., USA* (under review).
- Z.Chen; **H.Lang***; Y.Y.Xing. “Credibility-Guided Multi-Source Self-Training for Pavement Distress Detection.” *TRB 106th Annual Meeting of the Transportation Research Board, Washington, D.C., USA* (under review).
- Z.Zou; **H.Lang***; Y.Chen. “Strip Clustering-guided Semantic Segmentation for Pavement deformation Detection from MLS Point Clouds.” *Autom. Constr.* (under review).
- **H.Lang**; Z.Zou*; et al. “Robust Estimation for Pavement Plane Coefficients from Road Point Cloud.” *Autom. Constr.* (under review).
- **H.Lang**; J.Chen*; et al. “Active Learning for Infrastructure Distress Segmentation via Pixel-Level Difficulty Estimation.” *IEEE Trans. Intell. Transp. Syst.* (under review).

Peer-Reviewed Conference Papers

- [C17] X.T.Chen; **H.Lang*** (Presenter); Z.Zou; Y.Chen; Y.M.Jiang. “Coarse-to-refined Road Alignment Extraction and Parameterization from MLS Point Clouds.” *TRB 105th Annual Meeting of the Transportation Research Board*, Washington, D.C., USA, 2026 (Lectern).
- [C16] A.O.Sulaiman*; **H.Lang**; I.L.Al-Qadi. “Hot-Mix Asphalt Cracking Potential Prediction from Binder Rheological Parameters.” *TRB 105th Annual Meeting*, Washington, D.C., USA, 2026 (Lectern).
- [C15] **H.Lang***; W.D.Villamil; I.L.Al-Qadi. “3D Tire–Pavement Contact Stresses: Physics-Informed Prediction Approach.” *TRB 105th Annual Meeting*, Washington, D.C., USA, 2026.
- [C14] A.Singh*; **H.Lang**; I.L.Al-Qadi. “Seasonal Load Spectra on Illinois Interstate Highways.” *The 11th International Conference on the Bearing Capacity of Roads, Railways and Airfields*, Ljubljana, Slovenia, 2026.
- [C13] Y.M.Li; J.Chen; **H.Lang***; J.S.Qian. “Intelligent Pixel-Level Segmentation of Pavement Sealed Cracks Using CycleGAN-Based Domain Adaptation.” *TRB 104th Annual Meeting*, Washington, D.C., USA, 2025.
- [C12] Z.Chen (Presenter); **H.Lang**; Y.Y.Xing*; L.Wang; S.W.Zhang. “Aviation Accident Report Causality Extraction Based on Transformer with BERT-embeddings.” *TRB 104th Annual Meeting*, Washington, D.C., USA, 2025 (Lectern).
- [C11] J.Chen; Y.Yuan; **H.Lang***; S.Ding; J.J.Lu. “Efficient Data Selection Based on Difficulty-Aware Module for Pavement Distress Segmentation.” *TRB 103rd Annual Meeting*, Washington, D.C., USA, 2024.
- [C10] A.D.Wang; **H.Lang***; Z.Chen; Y.C.Peng; J.J.Lu. “The Two-step Method of Pavement Pothole and Raveling Detection and Segmentation Based on Deep Learning.” *TRB 103rd Annual Meeting*, Washington, D.C., USA, 2024.
- [C9] Z.Chen; **H.Lang***; A.D.Wang; Y.Peng; Y.Y.Xing. “Automatic Pavement Type Recognition for Image-based Data Using a Multi-Features Fusion Network.” *TRB 103rd Annual Meeting*, Washington, D.C., USA, 2024.
- [C8] Y.Peng; **H.Lang***; Z.Zou; J.J.Lu. “Road Curb Extraction from Unordered Point Clouds Based on Multi-Feature Filtering.” *TRB 103rd Annual Meeting*, Washington, D.C., USA, 2024.
- [C7] J.Chen; T.Wen; S.Ding; Y.Yuan; **H.Lang***; J.J.Lu. “Automatic Distress Segmentation on Asphalt Pavement Based on Deep Learning with Multiple Branches.” *TRB 101st Annual Meeting*, Washington, D.C., USA, 2022.
- [C6] A.D.Wang; S.Ding; **H.Lang***; T.Wen; J.J.Lu. “Rutting Abnormality Detection and Correction on 3D Asphalt Pavement Surfaces Using Segmentation Model.” *TRB 101st Annual Meeting*, Washington, D.C., USA, 2022.
- [C5] Y.Yuan; Q.W.Xue; **H.Lang***; J.Zhu; J.Chen; Y.Peng. “Smart Pavement: An Attention-Based Classification Model for Road Pavement Material.” *5th KES International Symposium on Smart Transportation Systems*, 133–140, 2022.
- [C4] S.Ding; Y.Y.Xing; **H.Lang**; T.Wen; J.J.Lu*. “Line-Structured Light Pavement Rut Detection with Markings Interference under Strong Light.” *TRB 99th Annual Meeting*, Washington, D.C., USA, 2020.
- [C3] **H.Lang**; Y.C.Peng; J.J.Lu*. “Automated Rutting Abnormality Detection on 3D Asphalt Pavement Surfaces Based on Disease Seeds.” *TRB 99th Annual Meeting*, Washington, D.C., USA, 2020.

- [C2] **H.Lang**; Y.X.Lou; J.J.Lu*. “Development of A Portable Pavement Surface Macrotexture Measurement System Applied in China.” *TRB 98th Annual Meeting*, Washington, D.C., USA, 2019.
- [C1] **H.Lang***; Y.X.Lou; J.J.Lu*; Y.Chen. “A Review of 3D Pavement Automatic Measurement System.” *17th COTA International Conference of Transportation Professionals*, Shanghai, China, 2018.

Patents, Standards & Books

Invention Patents (7)

- Road 3D Reconstruction Using Primary & Auxiliary LiDAR with Integrated Navigation. China Invention Patent (pending), 2024. **H.Lang**, Z.Zou, J.S.Qian, X.T.Chen.
- Road Roughness Detection with 3D Structured Light and Vibration Fusion. China Invention Patent (**authorized**), 2024. **H.Lang**, Z.Chen, J.S.Qian.
- Pavement 3D Detection Using a Bidirectional Triangulation System. China Invention Patent (**authorized**), ZL202210445208.9, 2023. **H.Lang**, Y.Yuan, J.Chen.
- Pavement Rutting Anomaly Detection System Based on 3D Triangulation. China Invention Patent (pending), 2023. J.M.Ling, **H.Lang**, J.S.Qian
- A Multidimensional Pavement Distress Data Processing Method. China Invention Patent (**authorized**), 2022. **H.Lang**, S.Ding, Y.Y.Xing.
- A 3-D Image Pavement Detection System. China Application Patent (**authorized**), CN 110415432 A, 2020. **H.Lang**, J.J.Lu.
- Communication Channel Extension System for IoT Gateway. China Invention Patent (**authorized**), ZL201610937604.8, 2019. R.Zhang, M.N.Guo, **H.Lang**, Y.Z.Peng, Y.Z.Wang.

Technical Reports

- I.L.Al-Qadi, M.Fakhreddine, **H.Lang**, M.Golparvar-Fard. *Computer Vision-Based Contactless Estimation of Asphalt Pavement Surface Friction*. FHWA-ICT-26-XXX (R27-247), 2026.
- I.L.Al-Qadi, J.J.García Mainier, **H.Lang**, L.Diab, G.Renshaw. *Optimizing the Use of Local Aggregates in SMA*. FHWA-ICT-26-004 (R27-216), 2026.
- I.L.Al-Qadi, A.O.Sulaiman, M.A.Uthman, **H.Lang**, J.J.García Mainier. *Using Advanced Binder Rheological Parameters to Predict Cracking Potential of Hot-Mix Asphalt Mixtures with Modified Binders* FHWA-ICT-25-011 (R27-250), 2025.

Standards & Books

- J.J.Lu, Y.M.Jiang, S.D.Chen, **H.Lang**, et al. *Automatic Bridgehead Settlement Detection System*. CHTS Group Standard (pending), 2024.
- J.J.Lu, Y.M.Jiang, R.J.Cao, M.Cao, **H.Lang**, et al. *Vehicle-Mounted 3-D Detection System for Pavement Distress*. CHTS Group Standard (authorized), 2023.
- J.J.Lu, Y.M.Jiang, **H.Lang**, S.D.Chen, Y.X.Lou. *Advanced Pavement Quality Evaluation and Detection Technology*. Tongji University Press, 2022.

Software & Open-Access Datasets

- “[ICT Optimal Mix Design Tool](#)” — AI-driven asphalt mix design web application deployed by ICT (R27-250).
- “[Asphalt Binder Selection Guidance Tool](#)” — Decision-support tool for IDOT binder selection (R27-250).
- **IP2RG Open Datasets** (github.com/IP2RG) — Asphalt Pavement Distress (18,256 samples), Pavement Type Classification (53,265 samples), Concrete Distress (7,414 samples), MLS Point Clouds (800 km, 9,056 samples). Updated 2025.

Professional Experience

Senior Research Scientist, Shanghai PRES Road Traffic Technology Co., Ltd 2021–2024

- Led R&D for vehicle-mounted 3-D pavement detection systems deployed on >50,000 km of Chinese national highways.
- Developed hardware-software integration for high-resolution 3-D cameras (AT-C6-4096) and LiDAR-INS fusion systems.

Research Scientist, Shanghai PRES Road Traffic Technology Co., Ltd 2017–2021

- Designed and calibrated 3-D imaging sensors for pavement inspection; contributed to 3 authorized patents.

Research Assistant, Tongji University (Advisor: Dr. Jian John Lu) 2017–2021

Undergraduate Research Assistant, Tianjin Univ. of Sci. & Tech. (Advisor: Dr. Fengzhi Dai) 2014–2016

Teaching

Courses Prepared to Teach: Highway & Geometric Design; Pavement Design and Management; Transportation Asset Management; Transportation Safety Analysis; Intelligent Transportation Systems; Sensing and Measurement for Civil Infrastructure; Computer Vision and 3D Reconstruction for Civil Systems; Machine Learning for Civil Engineering; Measurement and Control Circuits; Sensors; Principles of Automatic Control; Error Analysis, and Analog and Digital Circuits.

- **Pavement Design I CEE 406, Physics-informed Tire-Pavement Interaction** (Grainger College of Engineering, University of Illinois Urbana-Champaign. *(Guest Lecture)*)
- **Artificial intelligence II-LLM/VLM** (2027 FEB), Rantoul High School CS Visit, Illinois Center for Transportation. *(Co-Instructor, scheduled)*
- **STEP Teaching Training** (24-hr), Grainger College of Engineering, UIUC, Fall 2025–Spring 2026. *(Participant)*
- **Artificial intelligence** (2026 MAR), Rantoul High School CS Visit, Illinois Center for Transportation. *(Co-Instructor)*
- **Multimodal Infrastructure Intelligence: Measurement, Data, Method** (2-hr Workshop), CEE, UIUC, Fall 2025. *(Instructor)*
- **Smart Transportation & Digital Highway Workshop**, Tongji University, Spring 2024. *(Instructor)*
- **High-Quality Highway Construction Training**, Tongji University, Fall 2023. *(Instructor)*
- **Transportation Engineering – Intelligent Sensing and Detection Technologies**, Tongji University, Spring 2022. *(Co-Instructor)*

Mentoring

At UIUC (6 Ph.D. students, 1 M.S. student)

- Gaurav Bhusal (M.S. student, R27-288 IPFS project, 2026–)
- William Villamil Martinez (Ph.D. student, R27-277 WIM project, 2025–)
- AbdulGafar Sulaiman (Ph.D. student, R27-250 Binder project, 2024–2025)
- Aditya Singh (Ph.D. student, R27-277 WIM project, 2024–)
- Mohammad Fakhreddine (Ph.D. student, R27-247 Friction project, 2024–2026)
- Javier García Mainieri (Ph.D. student, R27-216 SMA-APT project, 2024–2025)
- Yihan Chen (Ph.D. student, FHWA GPR Matlab Tool, 2024–2026)

At Tongji University (8 graduate + 3 undergraduate students)

- Zheng Zou (Ph.D., LiDAR Fusion, 2020–2024) → **Assistant Professor, China Three Gorges University**

- Ye Yuan (Ph.D., 3D Reconstruction, 2020–2024) → **Assistant Professor, Nanjing University of Aeronautics and Astronautics**
- Aidi Wang (Ph.D., Infrastructure Monitoring, 2020–2024) → **Exchange Student, University of Cambridge**
- Junyi Hu (Undergraduate, 2023–2024) → **M.Sc. student, ETH Zurich**
- Yiming Li (Undergraduate, 2023–2024) → **Ph.D. student, Tongji University**
- Tian Wen (M.S.), Yuan Peng (M.S.), Xinting Chen (M.S.), Shuo Ding (Ph.D.), Jiang Chen (Ph.D.), Zhen Chen (Ph.D.), Li Chen (Undergraduate)

Invited Talks & Seminars

- 2026. *Road Alignment Extraction from MLS Point Clouds*. Road Profile Users' Group, Pittsburgh, PA.
- 2026. *Seasonal Load Spectra on Illinois Interstate Highways*. The 11th Int'l Conf. on the Bearing Capacity of Roads, Railways & Airfields, Ljubljana, Slovenia.
- 2026. *Coarse-to-refined Road Alignment Extraction and Parameterization from MLS Point Clouds*. TRB 105th, Washington, D.C.
- 2026. *HMA Cracking Potential Prediction from Binder Rheological Parameters*. TRB 105th, Washington, D.C.
- 2025. *Efficient Prediction of Asphalt Concrete Mix Design*. ASCE Int'l Airfield & Highway Pavements Conf., Glendale, AZ.
- 2025. *SCRIM Friction Number Prediction Using Decision Tree Algorithms*. Road Profile Users' Group, Dallas, TX.
- 2025. *Aviation Accident Report Causality Extraction*. TRB 104th, Washington, D.C.
- 2023. *Applications of Next-Gen AI in Highway Precision Inspection*. CHTS 5th Big Data Forum, Xi'an, China.
- 2022. *Smart Pavement: Attention-Based Classification Model*. KES-STs, Remote.
- 2021. *AI-Based 3D Pavement Inspection Technology*. CHTS 4th Big Data Forum, Guiyang, China.
- 2021. *Development and Application of Pavement Inspection Technologies*. World Transport Convention, Xi'an, China.

Honors & Awards

- 2026 — Scholarship and Teaching for Engineering Postdocs (STEP), Grainger College of Engineering, UIUC, (Awarded 800 USD)
- 2024 — First Prize, Shanghai Institute Traffic Engineering Science and Technology Award
- 2023 — Best Paper Award, 2nd CHTS Smart Transportation Annual Conference
- 2022 — Two Paper Awards (Technical Innovation & Technology Exploration), 12th CHTS Highway Maintenance Conference
- 2021 — Two Paper Awards, 11th CHTS Highway Maintenance Conference
- 2021 — Outstanding Graduate, Tongji University
- 2019 — First Prize (ranked 6th nationally among ~10,000 teams), China Graduate Mathematical Modeling Competition
- 2018 — Second Prize, China Graduate Mathematical Modeling Competition

Certifications

2026 — FAA Remote Pilot Certificate (Part 107) — Licensed for UAS field operations

Professional Service

Editorial & Committee Service

- Guest Editor, *Future Transportation*, Special Issue on Intelligent Transportation Infrastructure, 2026

- Guest Editor, *Journal of Transport Information and Safety*, Special Issue on AI-Driven Proactive Traffic Safety, 2024
- Visiting Research Scientist Search Committee, CEE, UIUC, 2024–2025
- Expert Committee Member, China Highway & Transportation Society (CHTS), 2024

Journal Reviewer

Transp. Res. Part C, Autom. Constr., IEEE Trans. Intell. Transp. Syst., Eng. Appl. Artif. Intell., Neurocomputing, Reliab. Eng. Syst. Saf., ASCE J. Comput. Civ. Eng., Int. J. Pavement Eng., Constr. Build. Mater., Struct. Control Health Monit., Measurement, and others.

Conference Reviewer

TRB, Int'l Conf. on Accelerated Pavement Testing (APT), ASCE ICTD, COTA CICTP.

Selected Media

- “Researchers develop stone-mastic asphalt using Illinois aggregate” UIUC/ICT, 2026.
- “Fakhreddine receives SCG Global Impact Prize at Cozad New Venture Challenge” UIUC/ICT, 2026.
- “ICT develops AI tool for optimal mix design of flexible roadways.” UIUC/ICT, 2025.
- “Cracking the code to predict modified asphalt binder performance.” UIUC/ICT, 2025.

Technical Skills

Programming: Python, C++ (13+ years), C#, VB, MATLAB, SQL

Machine Learning / Deep Learning: PyTorch, TensorFlow, scikit-learn

Computer Vision / Sensing: HALCON, 3D line-scan cameras (AT-C6-4096), LiDAR (DJI Livox), GNSS/INS (Bynav X1), IMUs, encoders, NI DAQ systems

Tools: Git, OpenClaw, LLM API, $\text{\LaTeX}$